Last time

Thm f cts. at a, $f'(x)$ exists $\forall x$ in an interval containing a, except maybe at $x=a$. Also, suppose that $\lim_{x\to a} f'(x)$ exists. Then $f'(a)$ also exists and $f'(a) = \lim_{x\to a} f'(x)$

Pf Strategy: Compute $f'(a)$ using limits

By definition, $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

We'll compute left + right limits separately.

$\lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0^+} f'(\alpha_h)$

By MVT, $\exists \alpha_h \in (a, a+h)$ s.t. $f'(\alpha_h) = \frac{f(a+h)-f(a)}{h} = \lim_{x\to a^+} f'(x)$

In considering $[a-h, a]$, we get $\lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a^-} f'(x)$

By assumption, we know $\lim_{x\to a} f'(x)$ exists, so $\lim_{x\to a^-} f'(x) = \lim_{x\to a^+} f'(x)$

So $f'(a)$ exists, & it is $\lim_{x\to a} f''(x)$.



Cauchy Mean Value Theorem

If f, g are cts. on $[a,b]$ & diff. on (a,b), then $\exists x \in (a,b)$ s.t. $(f(b)-f(a))\,g'(x) = (g(b)-g(a))\,f'(x)$

Rmk ① If $g(x) = x$, then you get MVT: $f(b)-f(a) = (b-a)f'(x)$

Rolle $\Rightarrow$ CMVT

This is equivalent to MVT.

② We can apply MVT separately to f & g, but we don't get CMVT. $\exists x_f \in (a,b)$ s.t.

$f'(x_f) = \frac{f(b)-f(a)}{b-a}$

$\exists\, x_g \in (a,b)$ s.t. $g'(x_g) = \frac{g(b)-g(a)}{b-a}$

So 2 applications of MVT don't guarantee that I get some x for both f & g.

Pf Let $h(x) = f(x)[g(b)-g(a)] - g(x)[f(b)-f(a)]$

$\exists\, x_0 \in (a,b)$ s.t. $h'(x_0) = \frac{h(b)-h(a)}{b-a}$

$$= \frac{\left(f(b)[g(b)-g(a)] - g(b)[f(b)-f(a)]\right) - \left(f(a)[g(b)-g(a)] - g(a)[f(b)-f(a)]\right)}{b-a}$$

$$= 0$$

But $h'(x) = f'(x)[g(b)-g(a)] - g'(x)[f(b)-f(a)]$

So at x_0, we have $h'(x_0) = 0$

$f'(x_0)[g(b)-g(a)] = g'(x_0)[f(b)-f(a)]$

∎

Have to check that we can apply MVT to $h(x)$

- $h(x)$ cts. on $[a,b]$

  f, g cts. on $[a,b]$, so $cf + dg$ cts. $\forall\, c, d$ constant
- For similar reasons, h is also diff. on (a,b) b/c f, g are diff.

∎

L'Hôpital's Rule

Suppose $\lim_{x\to a} f(x) = \lim_{x\to a} g(a) = 0$

& Suppose that $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists.

Then $\lim_{x\to a} \frac{f(x)}{g(x)}$ exists & is equal to $\lim_{x\to a} \frac{f'(x)}{g'(x)}$

Pf Strategy Apply CMVT to $[a, x]$. (also $[x, a]$)

If I were able to apply MVT

$\exists\, \alpha_x \in (a, x)$ s.t. $[f(x)-g(x)]g'(\alpha_x) = [g(x)-f(x)]f'(\alpha_x)$

Then we'd like to say that $\frac{f(x)}{g(x)} = \frac{f'(\alpha_x)}{g'(\alpha_x)}$

Now let $x \to a^+$:

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(\alpha_x)}{g'(\alpha_x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}$$

Same as $x \to a^-$

$$\lim_{x\to a^-} \frac{f(x)}{g(x)} = \lim_{x\to a^-} \frac{f'(\alpha_x)}{g'(\alpha_x)} = \lim_{x\to a^-} \frac{f'(x)}{g'(x)}$$

Check that CMVT can be applied.

① f, g cts. on $[a, x]$ & diff. on (a, x).

Given the assumption that $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists.

This says that f', g' exists near a.

f & g are both differentiable.

For continuity at x:

if f & g weren't cts. at some x

Replace by $x-\varepsilon$ for ε very small. Then they are diff. at $x-\varepsilon \Rightarrow$ cts. at $x-\varepsilon$.

We claim that f, g to be cts. at a by letting $f(a) = g(a) = 0$

Also, we have $\lim_{x \to a} f(a) = \lim_{x \to a} g(a) = 0$

The actual values of $f(a)$, $g(a)$ are irrevelant to L'Hôpital's Rule

- To check that dividing by $g'(x)$ is okay since $\lim_{x \to a} \frac{f'(x)}{g'(x)}$ is known to exist & $g'(x) \neq 0$ in an open property close enough to a, the quotient makes sense.

